

Kyuhwan Shim

Research Scientist
LG AI Research
150 Magokjungang-ro, Gangseo-gu
Seoul, Republic of Korea 07788

✉ kyuhwan.shim@snu.ac.kr
✉ kyuhwan.shim@lgresearch.ai
🌐 <https://bi.snu.ac.kr/khshim>
in [kyuhwan-shim](#)

RESEARCH EXPERIENCE

LG AI Research

Research Scientist, Physical Intelligence Lab
Research Scientist Intern, Physical Intelligence Lab

Aug 2026 – Present
Nov 2025 – Aug 2026

Universität Bonn, Deutschland

Visiting Student Researcher, Autonomous Intelligent Systems Group

Jul 2024 – Aug 2024

Seoul National University

Undergraduate Student Researcher, Biointelligence Lab
ExploreCSR, Google Research & Graduate School of Data Science

Dec 2023 – Aug 2024
Dec 2023 – Feb 2024

EDUCATION

Seoul National University, Seoul, Korea

Master Student in Graduate School of Artificial Intelligence

Sep 2024 – Present

Sogang University, Seoul, Korea

Bachelor of Engineering in Computer Science

Mar 2019 – Aug 2024

Chungnam Science High School, Chungnam, Korea

Mar 2016 – Feb 2019

WORK EXPERIENCE

TIDYBOY-DSPL, Seoul, Korea

Mar 2024 – Aug 2024

Vision Engineer

- Robocup2024 Team from Seoul National University, Pusan National University
- Developed Robot Vision Pipeline for real-time environment understanding

Nota Inc., Seoul, Korea

Mar 2023 – Aug 2023

Machine Learning Engineer Intern

- Developed Vision-based Pipeline Models for Intelligent Traffic System on Edge Device
- Construct Datalake and ML Pipeline on AWS and GCP

OCONNECT, Ulsan, Korea

Jun 2020 – Feb 2023

Co-Founder, Chief Design & Operating Officer

- **[LINK]** UNIST Based Student Venture which makes Easy-Insert Socket, OneStep
- R&D Didimdol, JAMES DYSON AWARD, Pre-startup Package, Youth Entrepreneurship Academy

Yeongdong Firestation, Yeongdong, Korea

Conscripted Firefighters (Military Duty)

Aug 2020 – Apr 2022

AWARDS &ACTIVITIES

Encouragement Award, Seoul National University

2024, 2025

Finalist (Top 3), Robocup Federation

Aug 2024

Finalist (Top 2), Seoul National University

2024

Top 30 & Talent Pool, Samsung Advanced Institute of Technology & DAICON

Sep 2023

Top 30, LG AI Research & Ministry of Employment and Labor

Sep 2023

Grand Prize, Gangseo-gu Office

2023

Grand Prize, National Information Society Agency, Ministry of Science and ICT

2022

Excellence Award, Chungnam Science Fair (64th)

2018

Special Prize, Chungnam Science Fair (63th)

2017

SCHOLARSHIPS

SNU AI Fellowship, Seoul National University

2024

Sogang University Alumni Scholarship, Sogang University Alumni Association

2022-2024

Priest Villarreal Memorial Smilegate Scholarship, General Scholarship, Startup Talents Scholarship

Albatross SW Scholarship, Sogang University

2019

MEDIA

Excellence Paper Award in the Undergraduate Division at KCC 2024, Sogang People

Jun 2024

'Nunettine' wins Grand Prize at Gangseo-gu Big Data Contest, Sogang People

May 2023

'DAVENGERS' wins BIGCONTEST Data Analytics Innovation League, Sogang People Dec 2022
UNIST venture 'OCONNECT' wins James Dyson Awards 2022, UNIST Press Sep 2022

PUBLICATIONS

INTERNATIONAL CONFERENCE

- [C4] **RAIN: Robotic Task Generalization with Region-Aware Interaction Networks**, Seunghun Lee*, **Kyuhwan Shim***, Yeogyong Song*, Seung-Hwan Kim, Sihaeng Lee, Byoung-Tak Zhang[†], Sunghoon Lim[†], Gunhee Lee[†], *X-Reason @ ECCVW 2026*,
- [C3] **Making Your Action Policies Interpretable: Mixture of Action Queries**, Suhyung Choi, Youngseok Joo, Hyundo Lee, Kisung Shin, **Kyuhwan Shim**, Chungwoo Lee, Minjeong Gu, Jun Ki Lee, Byoung-Tak Zhang[†], *Foundation Models Meet Embodied Agents @ CVPRW 2026, AI4GOOD @ ICMLW 2026*
- [C2] **ESPADA: Execution Speedup via Semantics Aware Demonstration Data Downsampling for Imitation Learning**, Byungju Kim, Pahk Jinu, ChungWoo Lee, Jaejoon Kim, Jangha Lee, Theo Taeyeong Kim, **Kyuhwan Shim**, Jun Ki Lee, Byoung-Tak Zhang[†], *IEEE Robotics and Automation Letters (RA-L) 2026, International Conference on Intelligent Robots and Systems (IROS) 2026*
- [C1] **CLIP-RT : Learning Language-Conditioned Robotic Policies from Natural Language Supervision**, Gi-Cheon Kang*, Junghyun Kim*, **Kyuhwan Shim**, Jun Ki Lee[†], Byoung-Tak Zhang[†] *Robotics: Science and Systems (RSS), 2025 (Oral)*

DOMESTIC JOURNALS

- [D4] **Scaling Language-Conditioned Policies with Fast and Lightweight Architectures**, Junghyun Kim*, Gi-Cheon Kang*, **Kyuhwan Shim**, Eungseo Kim, Jun Ki Lee[†], Byoung-Tak Zhang[†] *Korea Computer Congress 2024*
🏆 Excellent Presentation Award
- [D3] **Generalizing Language-Conditioned Robotic Policies via Large Language Models**, Junghyun Kim*, Gi-Cheon Kang*, **Kyuhwan Shim**, Eungseo Kim, Jun Ki Lee[†], Byoung-Tak Zhang[†] *Korea Computer Congress 2024*
🏆 Excellent Presentation Award
- [D2] **Stochastic Trajectory Diversification for Language-based Robot Action Learning**, Junghyun Kim*, Gi-Cheon Kang*, **Kyuhwan Shim**, Jun Ki Lee[†], Byoung-Tak Zhang[†] *Korea Software Congress 2024*
🏆 Excellent Presentation Award
- [D1] **KIRINO: An Interactive Chatbot System for User Persona**, Geon Choi*, Ganghun Kim*, Hyunjae Kim*, **Kyuhwan Shim***, Myoungwan Koo[†] *Korea Computer Congress 2024*
🏆 Excellent Paper Award

TEACHING EXPERIENCE

Artificial Intelligence (4190.408), Seoul National University Mar 2025 – Jun 2025
Translation Volunteer, Khan Academy & Chungnam Science High School Jun 2017 – Dec 2017

SPEECH

- **[PDF] Seminar** Samsung DICON Challenge - Domain Adaptation @ Release(Sogang CS) Dec 2023
- **[PDF] Seminar** Dive into YOLOv8 @ Nota Inc. Mar 2023
- **[PDF] Undergraduate Speech** at Carbon Neutral Workshop @ Shinhan Card & World Bank 2023

ENTREPRENEUR

OCONNECT - a student venture based at UNIST, founded to address usability challenges in everyday life. The team developed **ONE STEP**, a rotating, easy-to-plug-in electrical outlet designed to enable more intuitive and convenient plug insertion. insertion.

Awards

- **JAMES DYSON AWARD 2022 - Nat'l, Korea**, James Dyson Foundation
• Winner (TOP 3), International Competetion Jul 2022
- **PIN UP Concept Design Award 2021**, KAID
• BEST 100 in Living Goods Design Dec 2021

- **Busan-Ulsan-KyungPook Startup Day Innovation Ventures League**
Samsung Electronics
KyungPook, Ulsan, Busan Creative Econ. & Innov. Center
• Excellence Prize at Futures' League Dec 2021
- **Challenge K-Startup Expo : Collegiate Competition**
Ministry of SMEs and Startups,
Korean Institute of Startup & Entrepreneurship Development
• Top 10, Offline Exhibition Oct 2021
• Top 50, Offline Exhibition Aug 2019
- **Social Challenge alpha 7**, OhMyCompany, BSS, LISCC
• Challenge Award, Crowd Funding (Funded 1,424,000 KRW) Sep 2020 – Dec 2020
- **Ulsan Collegiate Startup Excellent Idea Contest**
Ulsan Creative Econ. & Innov. Center
• Winner : Grand Prize Nov 2020
- **Institute of Science & Technology Startups Competition**
KAIST, GIST, DGIST, UNIST, POSTECH
• Representative of UNIST (Top 2 at UNIST) Dec 2020
• Representative of UNIST (Top 3 at UNIST) Nov 2019

Fundings & Support Programs

- **DIDIMDOL First Step - Startup Growth Development Project**,
Ministry of SMEs and Startups,
• Recommended from Ulsan Creative Econ. & Innov. Center Aug 2022 – Jul 2023
- **Ulju Youth Startup Academy**, Ulsan Youth Startup Center Mar 2022 – Dec 2022
- **Pre-IBK Changgong @ UNIST**, UNIST, IBK Changgong Jul 2022 – Nov 2022
- **Private-Sector Open Start-up Revitalization Project**,
Ulsan Creative Econ. & Innov. Center May 2022 – Oct 2022
- **BUSAN BIGS Program**,
Busan-si, KSD, KHUG, KOSPO, HF, KIBO, Busan PA, BNK Mar 2022 – Sep 2022
- **Innopolis Campus for Ulsan Collegiate**, UNIST, Innopolis Jun 2021 – Aug 2022
- **12th UEPA Startup Incubator**, Ulsan Youth Startup Center Mar 2021 – Dec 2021
- **SBC Youth Entrepreneurship Academy**, Korea SMEs and Startups Agency
• Most Funded Startup Company (100,000,000 KRW)
• Excellent Completion at Pre-School
• 'Pre-School' Completion of Excellence & Youth Startup Academy Nov 2020 – Nov 2021
- **UNIST UNicorn Start-Up Support Program**, UNIST
• Funding (12,000,000 KRW)
• Participated for Four Semesters Sep 2019 – Feb 2021
- **KISED Pre-Startup Package (General Field)**,
Korea Institute of Startup & Entrepreneurship Development
• Most Funded (70,000,000 Won) Apr 2020– Dec 2020

Media Coverage

- **UNIST Startup OCONNECT wins at JAMES DYSON AWARD 2022**,
UNIST Press, CHOSUN Ilbo, Segye Ilbo, Sisa-news, Newsis Feb 2022
- **UNIST 2021 Innopolis Performance Presentation**, UNIST Press, UlsanDaily, ETNews Feb 2022
- **UNIST UNISPARK for Young Entrepreneurs** UNIST Magazine 2021 Winter Dec 2021
- **Southeast Venture Competition held by UNIST Lab Startup Innovation Group**
The Korea Economy TV Dec 2021
- **"New evolution of electrical outlets..." OneStep is released** The Korea Economic Daily Dec 2021
- **"Make a 'National Outlet' which 'Connects' warmth to each other"** UNIST Magazine 2021 Spring Feb 2021
- **"Doing businesses regardless of COVID-19" - Leap forward with technology** SBS, UBC Feb 2021

